STAFF REVIEW · H446 222 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. No. 7 is an overestimate.

2. 7.

3. Non-negative edge weights.

4. 0–1 and 3–4.

I Do — source worked steps

1. Predict first — Start with pivot 4; 5 is too large to move to the smaller section.

2. Model response — Move 2 to the left: [2,5,4].

3. Model response — Swap pivot into index 1: [2,4,5]. Only one partition is being demonstrated.

We Do — guided CHECK

1. Model response
From Q, R improves to 3 and T gets 10.
From R, T improves to 6. The shortest P→T path is P→Q→R→T, total cost 6.

You Do — source mark schemes

1. MARK SCHEME /10: pivot 5 [1]; after moving 2 [2,7,6,1,5] [1], after moving 1 [2,1,6,7,5] [2] (no further value qualifies); pivot placed [2,1,5,7,6] [1], pivot index 2 [1]; fully sorted [1,2,5,6,7] [2]; average O(n log n) [1], worst $O(n^2)$ [1].

2. MARK SCHEME /10: after A, B2 and C6 [2]; settle B2, improve C3 and set D9 [2]; settle C3, improve D5 and set E12 [2]; settle D5, improve E8 [2]; shortest route A→B→C→D→E [1] costs 2+1+2+3=8 [1].

3. MARK SCHEME /10: g=cost travelled [1], h=estimated remaining cost [1], f=g+h [1]; after S, f(A)=2+3=5 [1], f(B)=5+2=7 [1]; after A, B improves to g3 [1], f(B)=3+2=5 [1]; S→A→B→G [1] cost 2+1+3=6 [1]; h(A)=3 is admissible because true remaining cost is 4 [1].

Exit Ticket — source answer

1. False. They address different problems: ordering values, shortest paths and heuristic-guided search respectively.